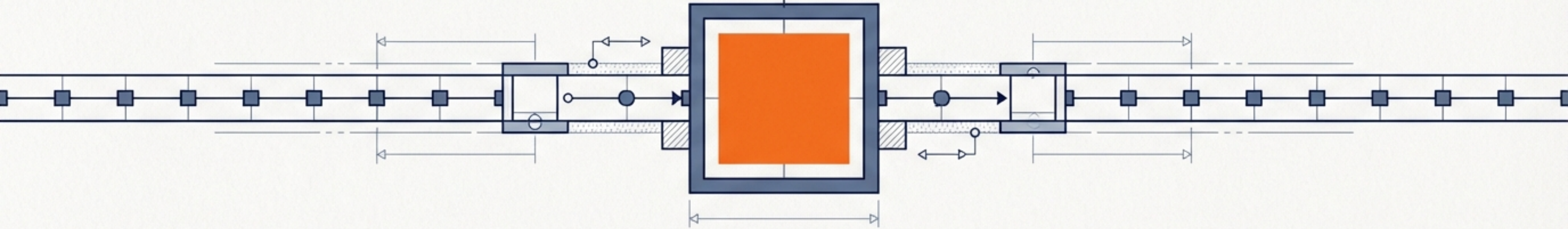
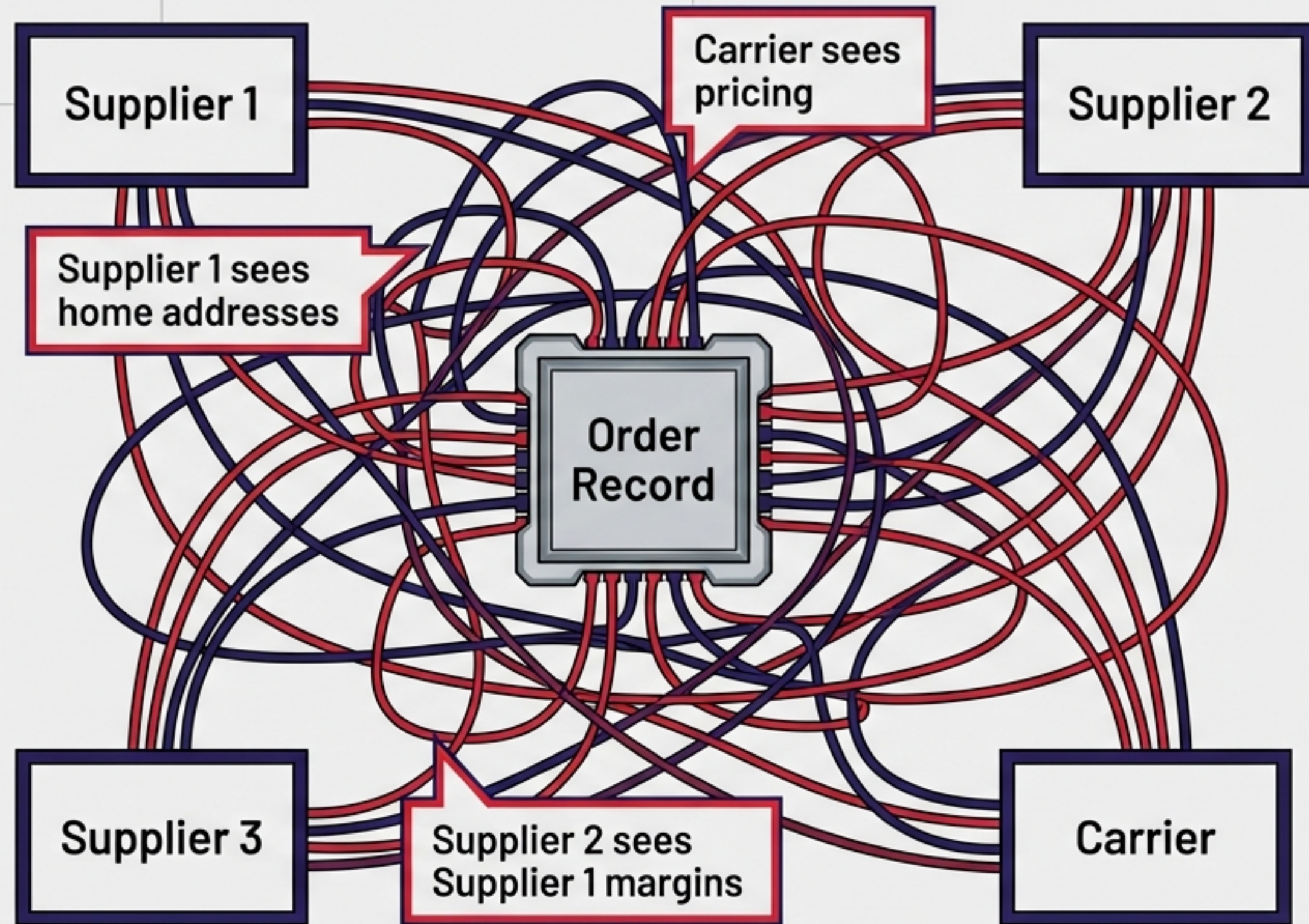


The Cryptographic Assembly Line

Enforcing supply chain data minimisation
through physics, not policy. // v0.16.17



The reality of the modern supply chain: everyone sees everything.



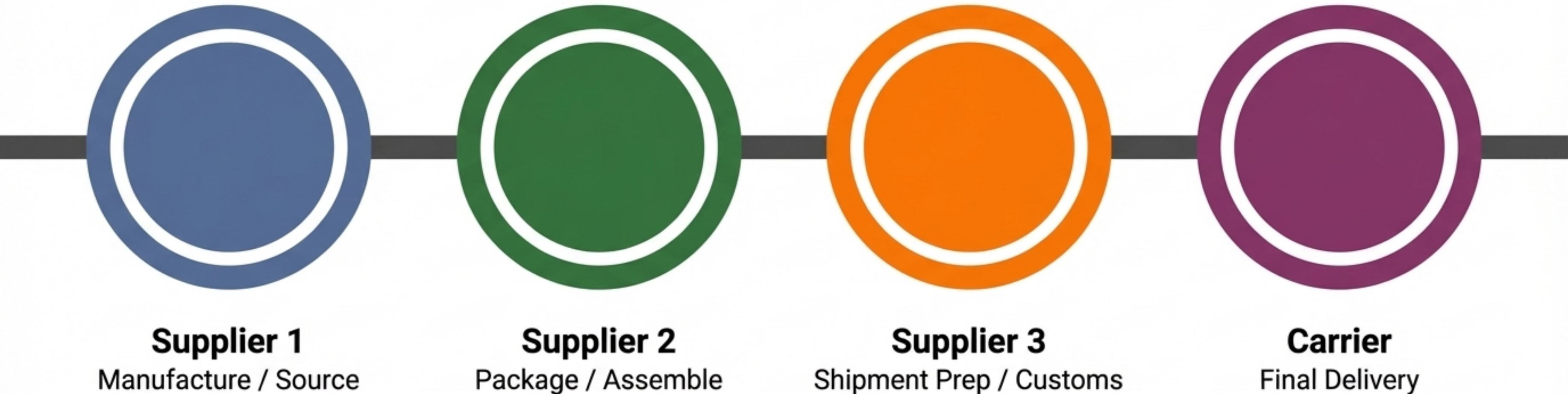
Today's fulfilment relies on emails, ERP systems, and shared spreadsheets.

Data minimisation is purely aspirational.

We rely on policies and Access Control Lists (ACLs) to protect sensitive data, but structurally, the data is over-exposed by default.

A single order. Four independent actors. Zero shared trust.

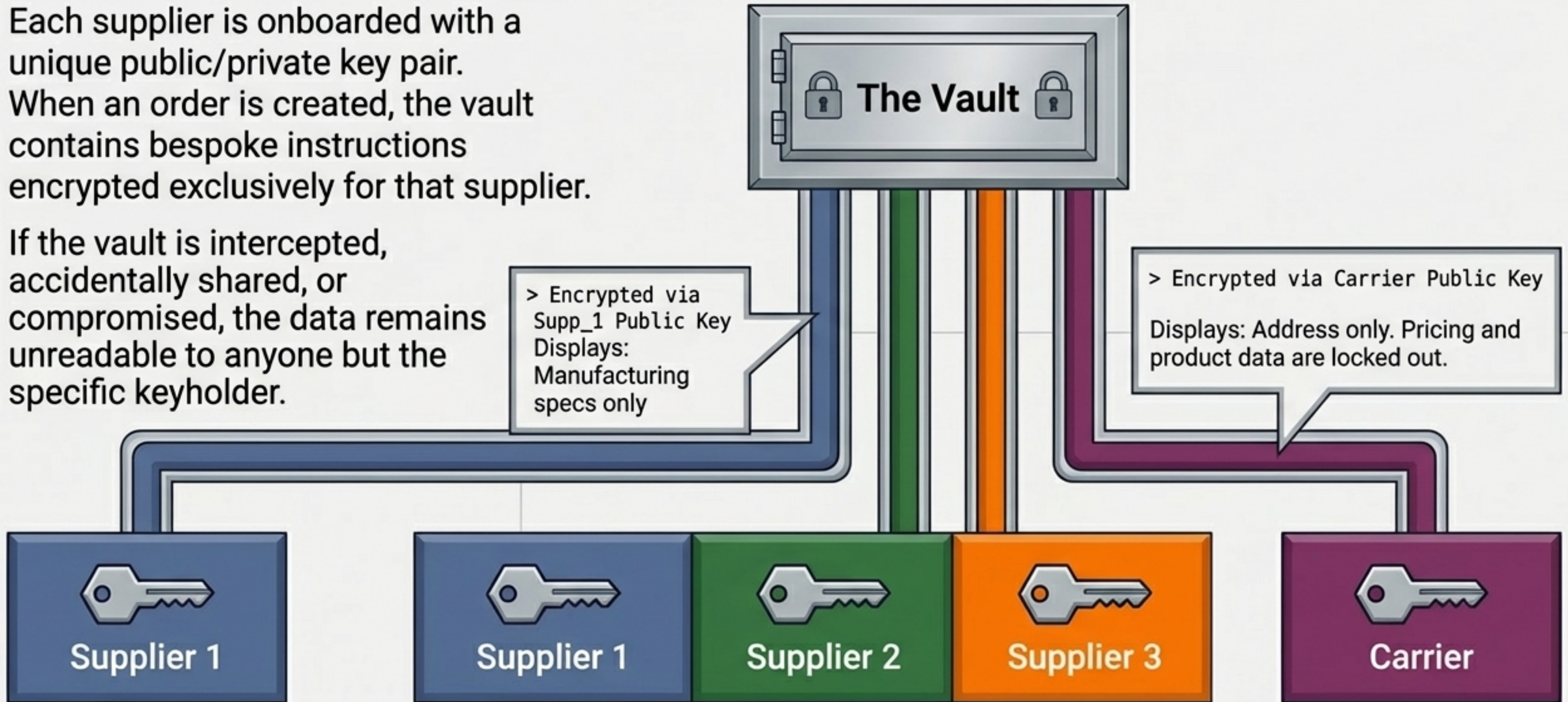
Fulfilling an order requires independent entities to coordinate. The carrier requires the delivery address, but not the manufacturing specifications. Supplier 1 requires the manufacturing specifications, but not the customer's home address.



Per-entity encryption enforces data minimisation by design.

Each supplier is onboarded with a unique public/private key pair. When an order is created, the vault contains bespoke instructions encrypted exclusively for that supplier.

If the vault is intercepted, accidentally shared, or compromised, the data remains unreadable to anyone but the specific keyholder.



Supply chains are sequential. Cryptography must be, too.

Supplier 2 cannot physically act before Supplier 1 finishes.

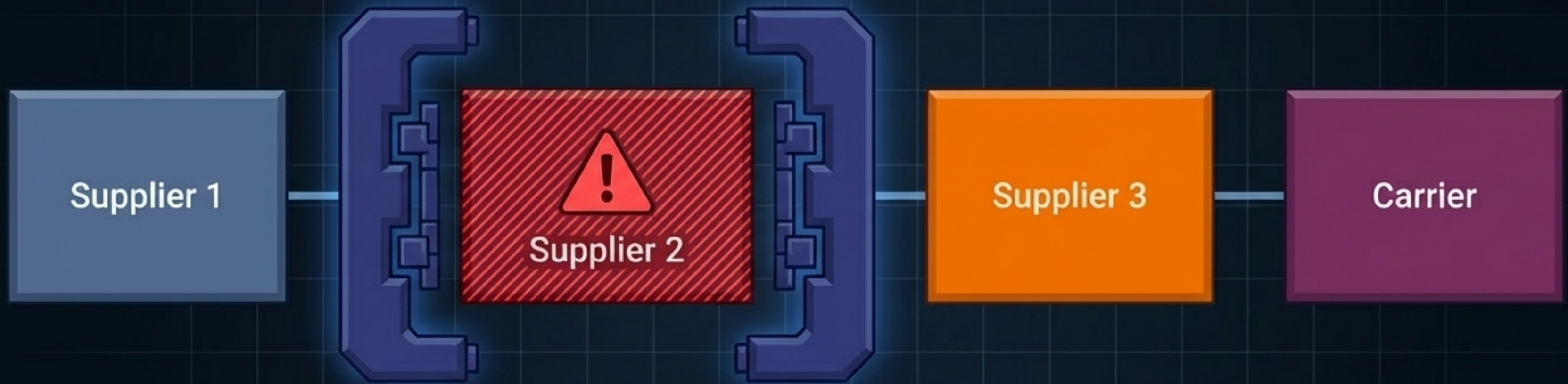
The Vault mirrors this reality through signed handoffs.

Supplier 2's instructions are not protected by a mere policy check; the handoff message literally does not exist for Supplier 2 to read until Supplier 1 creates and signs it.



Compromise containment: restricting the blast radius.

If a node is compromised, the attacker accesses only the instructions encrypted for that specific node and the handoff messages it has received.



Exposed

Supplier 2 packaging instructions.

Contained (Safe)

Supplier 1 data, Supplier 3 customs data, Customer delivery address.

Shifting from Access Control Lists to Cryptographic Physics.

The Status Quo ❌

Enforcement: Policy-based (ACLs)

Data Visibility: Over-exposed
(everyone sees everything)

Workflow: Independent of security

Breach Impact: System-wide cascade

The Vault Model ✅

Enforcement: Cryptographic (Math)

Data Visibility: Strict Need-to-know

Workflow: Bound to sequential signed handoffs

Breach Impact: Compartmentalised
to a single node

Natural regulatory alignment for modern supply chains.

1. Data Minimisation:

Each supplier accesses only relevant data.

5. Right to Erasure:

Deleting the specific vault key renders the data permanently unreadable.

4. Provenance:

Every action is cryptographically signed, signed, creating an immutable audit trail.



2. Purpose Limitation:

Encryption scope strictly matches the entity's functional purpose.

3. Storage Independence:

Encrypted data can be stored on any server globally; the server cannot read the data.

Two deployment models. Matching your supply chain maturity.



Zip-Based (Simple)

- **Mechanism:** Vault exported as a zip, sent via email, WhatsApp, or existing channels.
- **Action:** Supplier opens zip, enters private key.
- **Advantage:** Matches today's reality. Zero server dependency. No specialised accounts required.



Server-Based (Real-Time)

- **Mechanism:** Vault lives on SG/Send infrastructure.
- **Action:** Accessed via transaction ID and private key.
- **Advantage:** Real-time status updates. Centralised controller visibility. Suppliers push completion messages directly to the vault.

The breakthrough: Cryptographic State Management.



The true power of the Vault model is not merely secure file transfer. Because data for the next step literally does not exist until the previous step signs off, the security model is the **workflow model**. State is managed by **cryptographic enforcement**, **eliminating the gap** between what a system is supposed to do and what it can do.

The foundation for a new ecosystem.

This pattern is more than a tool; it is a standalone company waiting to be built.

SG/Send provides the raw primitives: encrypted file transfer, per-entity encryption, signed sequential handoffs, and **vault-based** state management.

The **opportunity** lies in building the domain-specific workflow UI, industry relationships, and supply chain expertise on top of this unbreakable cryptographic foundation.

